A REDUCED LOCALLY FINITE TYPE SCHEME OVER A FIELD IS GEOMETRICALLY REDUCED EXACTLY WHEN IT CONTAINS A DENSE SMOOTH OPEN

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is complete as written and ends with the existence of a smooth principal localization; no step is referred to the reader.

Editorial note. The counted spans carry 5 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-geometrically-reduced-dense-smooth-open is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Smoothness implies geometric normality and geometric reducedness. Same chapter, Lemma lemma-smooth-geometrically-normal, invoked at proof lines 4704-4705. Supplies that the dense smooth open is geometrically reduced at the generic points.
- Geometric reducedness can be tested at generic points. Same chapter, Lemma lemma-generic-points-geometrically-reduced, invoked at proof lines 4706-4707. Supplies the passage from generic points to the whole scheme.
- A geometrically reduced field extension of a field is separable. Algebra, Lemma algebra-lemma-characterize-separable-field-extensions, invoked at proof lines 4719-4720. Supplies separability of the function field over k.
- Separable residue field implies smoothness at a prime for finite type algebras. Algebra, Lemma algebra-lemma-separable-smooth, invoked at proof lines 4722-4724. Supplies that the affine chart is smooth at the zero ideal, hence that a principal localization is smooth.

Source reference. Original lemma 030W: sources/stacks-chow/tags.txt, source lines 3963–3963. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 00TV: sources/stacks-chow/tags.txt, source lines 1089–1089. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 04KS: sources/stacks-varieties/pinned-original-source.tex, source lines 555–555. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 056T: sources/stacks-varieties/pinned-original-source.tex, source lines 4617–4617. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01ON: sources/stacks-chow/tags.txt, source lines 2200–2200. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let X be a reduced scheme that is locally of finite type over a field k. Then X is geometrically reduced over k if and only if X contains a dense open which is smooth over k.*

Proof. The problem is local on X, hence we may assume X is quasi-compact. Let $X = X_1 \cup \ldots \cup X_n$ be the irreducible components of X. First suppose that X contains a dense open which is smooth over k. Then X is geometrically reduced at the generic point of each X_i (for example by Lemma 056T). By Lemma 04KS, it follows that X is geometrically reduced.

Conversely, suppose that X is geometrically reduced. Since $Z = \bigcup_{i \neq j} X_i \cap X_j$ is nowhere dense in X, we may replace X by $X \setminus Z$. As $X \setminus Z$ is a disjoint union of irreducible schemes, this reduces us to the case where X is irreducible. As X is irreducible and reduced, it is integral, see Properties, Lemma 01ON. Let $\eta \in X$ be its generic point. Then the function field $K = k(X) = \kappa(\eta) = \mathcal{O}_{X,\eta}$ is geometrically reduced over k, hence separable over k, see Algebra, Lemma 030W. Let $U = \mathrm{Spec}(A) \subset X$ be any nonempty affine open so that $K = A_{(0)}$ is the fraction field of A. Apply Algebra, Lemma 00TV to conclude that A is smooth at (0) over k. By definition this means that some principal localization of A is smooth over k and we win. $\square$